INDIA INNOVATES 2026

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GeoSynapse

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PROBLEM STATEMENT

CORE PROBLEM

- DIGITAL DEMOCRACY

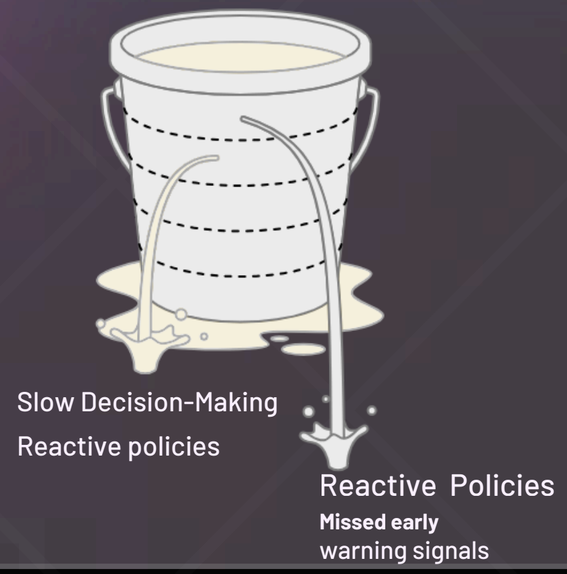
Critical global signals exist, but their relationships remain invisible.

- Large volumes of data across geopolitics, economics, defence, technology, climate, and society are generated daily, but they remain fragmented across reports, datasets, and systems, making connections between events difficult to understand.
- There is no unified intelligence system that can combine structured data, unstructured information, and real-time feeds, limiting the ability to generate timely and meaningful insights.

Why This Problem Matters?

- Global events are highly interconnected and directly affect India's strategy.
- Agencies often work with separate data and analyses.
- This creates fragmented understanding, causing slow decisions and policy delays.
- The problem is not data shortage, but the lack of a unified system to connect and interpret it.

Data Overload, Intelligence Deficit

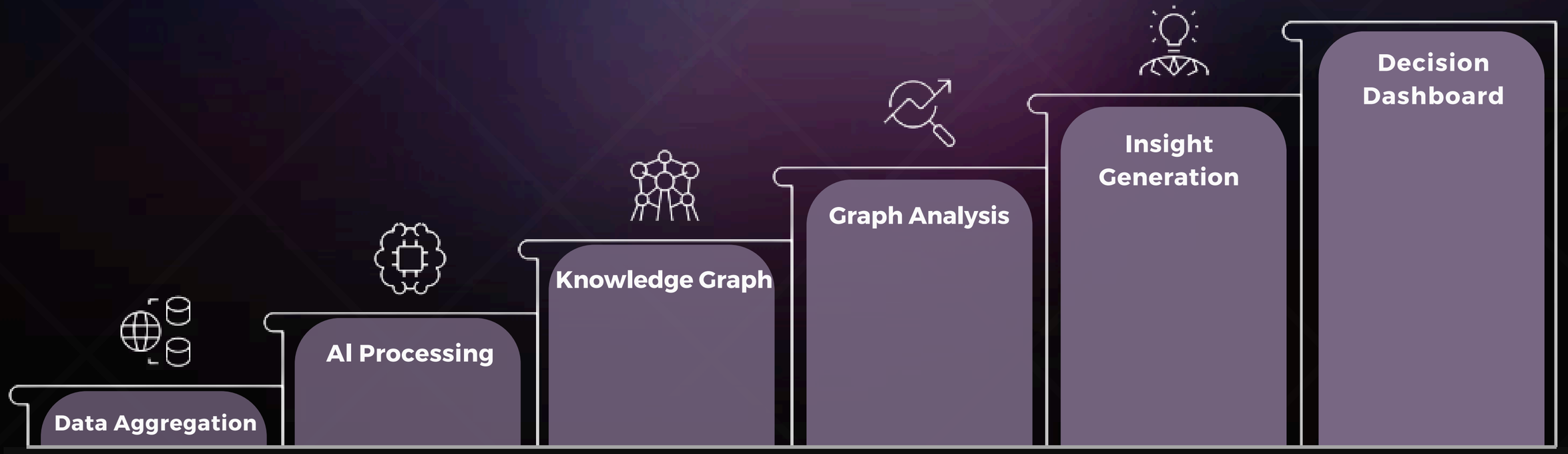


SOLUTION

Global Ontology Engine (GOE)

An AI-powered intelligence platform that integrates global data from multiple domains into a unified knowledge graph for strategic decision-making.

- Transforms scattered information into connected insights and risk indicators
- Enables policymakers and analysts to make faster, explainable, data-driven decisions.



KEY DIFFERENCES & IMPACT

Impact

- Faster strategic decisions using real-time intelligence
- Early risk detection for supply chains, geopolitics
- Data-driven policy planning instead of manual analysis
- Clear visibility of global dependencies across sectors

Practicality

- Uses existing open data sources
- Proven technologies: NLP, knowledge graphs, dashboards
- Scalable architecture that grows over the time
- Useful for government policy and strategic research

- Demo link is attached on last page

Existing Gap	Our solution
• Data scattered across multiple silos	• Unified intelligence graph connecting global data
• Mostly single domain analysis	• Cross-domain intelligence analysis
• Requires manual analysis of multiple reports	• AI-driven relationship discovery
• Works on static or batch datasets	• Real-time global data integration
• Limited transparency in insights	• Explainable insights with traceable sources

ARCHITECTURE

Architecture Framework

Decision Dashboard
Present insights through interactive visualizations

Insight Generation
Produce explainable intelligence and risk scores

Graph Analysis
Identify dependencies and cascading impacts

Knowledge Graph
Organize information into an ontology-driven graph

AI Processing
Extract entities and relationships using NLP

Data Aggregation
Collect global data from various sources

System Architecture Flow

Global Data Sources
News, OSINT, finance, sensors,social media, etc.

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Data Ingestion & Integration

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Real-Time Processing

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Knowledge Graph Construction

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AI Reasoning & Risk Analysis

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Decision Intelligence Dashboard

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Policy Makers / Analysts

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TECHNOLOGY USED

Data & Graph Layer
- Neo4j – knowledge graph database
- ChromaDB / FAISS – vector store

AI & NLP
- OpenAI GPT+LangChain- GraphRAG reasoning
- spaCy / BERT – entity & relation extraction

Backend
- Python + FastAPI
- APIs and processing

Frontend
- HTML / CSS / JS
- D3.js – graph visualization

Infrastructure
- Docker / Docker Compose – deployment
- Redis -caching

Neo4j Graph Database

OpenAI GPT & LangChain

FastAPI Web Framework

ChromaDB & FAISS

Docker & Docker Compose

D3.js Visualization

BERT & spaCy

HTML, CSS, JavaScript

Redis Cache

NewsAPI Integration

Backend Technologies

Frontend Technologies

FEATURE / USP

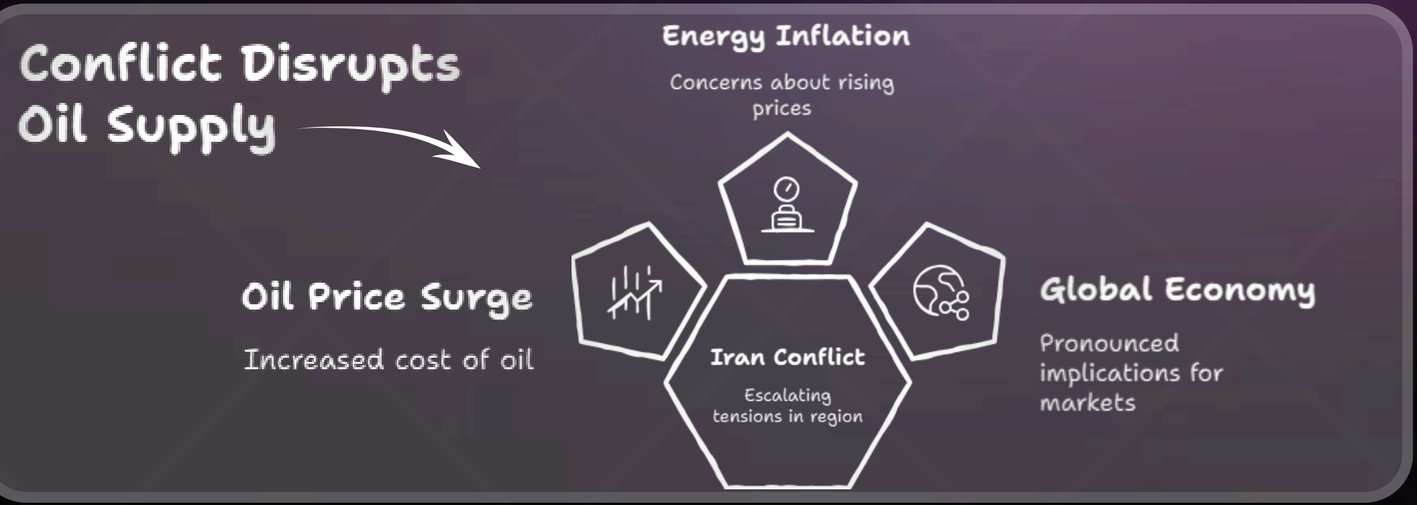
FEATURE

AI-Powered Explanations feature
Generates clear explanations with supporting evidence, showing how each insight or conclusion was derived.

Strategic Risk Scoring usp
Assigns risk scores to potential threats like supply-chain disruptions or geopolitical tensions to help prioritize decisions.

Unified Strategic Dashboard
Provides a single interface to view event alerts, risk indicators, and intelligence graphs.

Custom Filtering & Regional Analysis
Allows analysis by region, domain, or time period for focused strategic insights.



Cross-Domain Intelligence Platform integrating **multiple global sectors in a single system.**

Graph-based with explainable AI insights, not just raw analytics.

Natural Language Query Interface (LLM) allowing users to ask strategic questions and receive data-backed answers.

Real-time intelligence generation instead of static reports or delayed analysis.

REFERENCES / LINKS

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- **Yao, L. et al. (2025)**. “From knowledge graph construction to retrieval-augmented generation: a framework for comprehensive earthquake emergency support.” Geo-spatial Inf. Sci. (Jul 2025) (Hybrid KG+RAG system for disaster management).
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- **Hunger, M. (2024).** “What Is GraphRAG?” Neo4j Blog (Dec 2024) (Explains GraphRAG as a graph-powered retrieval layer for GenAI).
- **Collarana, D. et al. (2026).** “Graph RAG in the Wild: Insights and Best Practices from Real-World Applications.” Semantic Web Journal (2026) (Case studies of Graph RAG across industries, introduces FRAG-KEDA engine).
- **Fan, W. et al. (2025).** “MedRAG: Enhancing RAG with Knowledge Graph-Elicited Reasoning for Healthcare Copilot.” Proc. ACM Web Conf 2025 (KG-augmented RAG for medical QA).
- **IBM (2024).** “What Is GraphRAG?” IBM Think Blog (Nov 2024) (Defines GraphRAG as RAG augmented with knowledge graphs).
- **NVIDIA (2025).** “What Is Retrieval-Augmented Generation (RAG)?” NVIDIA Blog (Jan 2025) (Overview of RAG technique for grounding LLMs in external data).

Demo Video : https://youtu.be/aUDoY588tow?si=NyJ-WSF-39QFIkyV

GitHub url : https://github.com/Vagishaa11/GeoSynapse

THANK YOU